Development of a Deep Learning-based System for Sign Language Recognition from Hand Gestures

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Abstract

Bangla Sign Language (BdSL) lacks extensive open-source, word-level datasets and multimodal architectures, limiting recognition accuracy and vocabulary coverage. This research introduces a 300-word BdSL recognition system combining existing and newly collected high-quality video data from diverse signers. A multimodal mid-stage fusion model integrates RGB spatio-temporal features with kinematic landmarks of hands, face, and arms, allowing the system to jointly learn visual and motion cues [1]. This fusion model significantly enhances robustness and accuracy, establishing a scalable and effective framework for real-world BdSL communication and practical application.

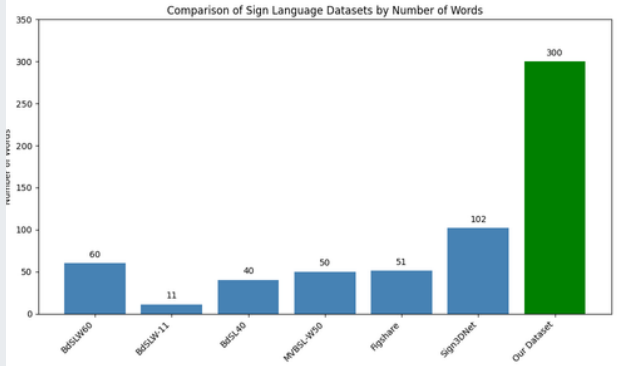
Objectives

- Develop comprehensive word-level BdSL dataset.
- Create accurate deep learning recognition model.
- Implement multimodal fusion approach.
- Enable real-world practical application, which is lacking in Bangladesh [2]
- Establish scalable framework for BdSL recognition.

Data Collection

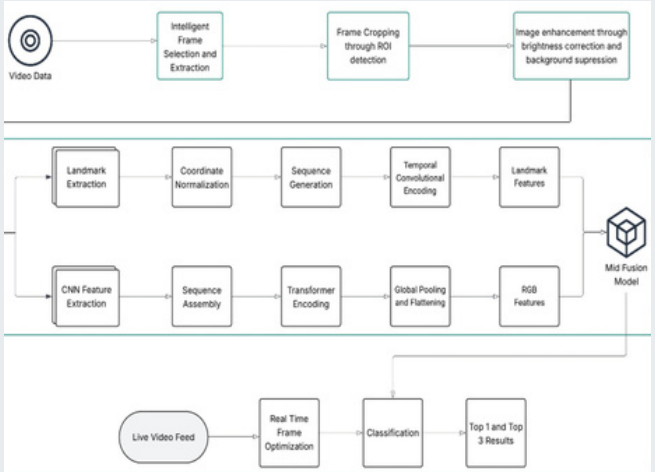
- Collected 100 everyday usable words
- High-quality video data from diverse signers
- Creates comprehensive lexicon for daily use
- Largest open-source BdSL dictionary to date
- Ensures linguistic breadth and diversity

Comparison of Dataset

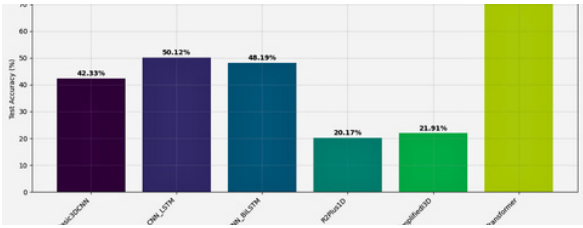


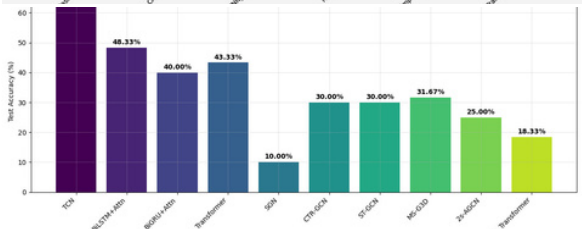
Methodology

- RGB Stream: CNN Transformer architecture captures spatio-temporal dynamics
- Landmark Stream: TCN models kinematic motion cues (hands, face, arms)
- Mid-Level Fusion: Integrates RGB and Landmark features for robust recognition



Results and Analysis





- CNN_Transformer stream excels at spatial feature extraction
- TCN excels at modelling landmark features.
- Combined approach achieves highest accuracy

Conclusion

The mid-level combination of Landmark and RGB models improves recognition performance and resilience. Hyperparameter tuning, knowledge distillation, pruning, quantization, and attention mechanisms will further enhance performance in future work. Our 100-word extension, creates the largest publicly available BdSL dictionary and presents a scalable, unbiased, and high-performance system for real-world recognition.

References

[1] Gökçe, Ç., Özdemir, O., Kındıroğlu, A. A., & Akarun, L. (2020). Score-level multi cue fusion for sign language recognition.
[2] Rahaman, M. A., Mahmud, M., & Kabir, A. (2018). Bangla language modeling algorithm for automatic recognition of hand-sign-spelled Bangla sign language. Frontiers of Computer Science, 12(2), 277-289.